



Biology of Blood and Marrow Transplantation

journal homepage: www.bbmt.org

ASBMT
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Impact of the Use and Type of Antibiotics on Acute Graft-versus-Host Disease



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Article history:

Received 29 March 2018

Accepted 26 June 2018

Key Words:

Intestinal microbiota
Acute graft-versus-host disease
Fourth-generation cephalosporins
Allogeneic hematopoietic stem cell transplantation

A B S T R A C T

The intestinal microbiota plays an important role in the pathogenesis of acute graft-versus-host disease (aGVHD). During the course of hematopoietic stem cell transplantation (HSCT), the intestinal microbiota is influenced by the use of broad-spectrum antibiotics. However, the impact of the use and type of antibiotics on the microbiota composition and, subsequently, the onset of aGVHD remain poorly understood. We hypothesized that the use and type of antibiotics had an impact on the occurrence of aGVHD. We assessed 275 patients who underwent their first allogeneic HSCT between January 2005 and June 2015 at Kyoto University Hospital. We monitored the 6 most frequently administered antibiotics (fourth-generation cephalosporins, glycopeptides, piperacillin-tazobactam, carbapenems, aminoglycosides, and quinolones) administered between days −14 and +14 relative to HSCT and its duration. The primary endpoint was the cumulative incidence of grades II to IV aGVHD. The cumulative incidence of aGVHD was significantly higher in patients administered fourth-generation cephalosporins than in patients not receiving fourth-generation cephalosporins (grades II to IV: hazard ratio, 1.98; 95% confidence interval, 1.19 to 3.29; $P = .0087$; grades III to IV: hazard ratio, 8.03; 95% confidence interval, 1.07 to 60.51; $P = .043$). In contrast, there was no significant association between administration of other antibiotics and aGVHD incidence. As for organ-specific aGVHD, the cumulative incidence of gut aGVHD was significantly higher in patients who received fourth-generation cephalosporins than in those who did not (31% versus 16%, $P = .018$). In conclusion, we demonstrated that the administration of fourth-generation cephalosporins had a strong impact on the development of aGVHD.

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INTRODUCTION

Acute graft-versus-host disease (aGVHD) and infection remain significant life-threatening complications after hematopoietic stem cell transplantation (HSCT). Most patients require treatment with broad-spectrum antibiotics for neutropenia and mucosal injury after conditioning regimens. The use of broad-spectrum antibiotics may also influence the intestinal microbiota. However, the effect on the microbiota composition and, subsequently, systemic homeostasis remain poorly understood.

Recent advances have highlighted the interaction between the intestinal microbiota and innate and adaptive immunity [1,2]. Previous studies have reported the association between the composition of intestinal microbiota and autoimmune disorders and inflammatory bowel disease [3,4]. In the field of HSCT, the intestinal microbiota was identified as an important risk factor for aGVHD onset in seminal experiments performed in the 1970s [5].

Based on these findings and early preclinical studies, many HSCT programs have introduced gut decontamination, the use of which, however, is controversial [6,7]. In recent years microbiota abundance has been recognized as a more protective factor than the growth suppression of microbiota and Taur et al. [8] reported that the diversity of intestinal microbiota at the time of stem cell engraftment improved allogeneic HSCT outcomes.

During neutropenia after HSCT, patients are often exposed to large amounts and various types of antibiotics. This leads to the profound modification and disruption of the intestinal microbiota. Therefore, the use of antibiotics around the time of HSCT may influence immune response and subsequently the occurrence of aGVHD through intestinal microbiota alterations. Thus, we hypothesized that the use and type of antibiotics impact the occurrence of aGVHD and performed a retrospective analysis.

METHODS

Study Design

The medical records of 275 consecutive adult patients with hematologic disorders who underwent their first allogeneic HSCT between January 2005 and June 2015 at Kyoto University

Financial disclosure: See Acknowledgments on page 2183.

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<https://doi.org/10.1016/j.bbmt.2018.06.031>

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Hospital were reviewed. Patients who had active infections at day –14 from the transplantation procedure and those who had developed engraftment failure after the transplantation were excluded; thus, a total of 241 patients were included in the analysis, without any duplication of subjects. This study was approved by the Ethics Committee of Kyoto University Graduate School as well as the Faculty of Medicine. Written informed consent for the transplantation protocol was obtained from all patients.

Monitoring the Use of Antibiotics and Diagnosis of GVHD

We monitored the 6 most frequently administered antibiotics (fourth-generation cephalosporins, glycopeptides, piperacillin-tazobactam, carbapenems, aminoglycosides, and quinolones) administered between days –14 and +14 relative to allo-HSCT, as well as their administration duration. aGVHD was diagnosed clinically, certified pathologically whenever possible, and classified according to standard criteria [9]. Late-onset aGVHD [10] was included in the definition of aGVHD.

Guideline for the Initiation and Prophylactic Prescription of Antibiotics

At our institution, empirical broad-spectrum antibacterial therapy is started immediately after 2 sets of blood cultures are obtained in all febrile neutropenic patients. Fourth-generation cephalosporins have been the standard first-line antibiotics for the treatment of febrile neutropenia. Piperacillin-tazobactam is used as an alternative first-line treatment if patients have a history of bacterial infection resistance to fourth-generation cephalosporins, allergy to fourth-generation cephalosporins, or preference of the physicians in charge. Second-line antibiotics (eg, carbapenems and aminoglycosides) are used when fever is prolonged despite the initiation of first-line antibiotics. Since 2003, levofloxacin prophylaxis in patients undergoing allogeneic HSCT has not been routinely prescribed at our institution. We use quinolones prophylaxis in the situation of the history of bacterial infection sensitive to quinolones and so on.

Statistical Analysis

The primary endpoint was the cumulative incidence of grades II to IV aGVHD after day 14 from HSCT. The landmark day was set as day 14. To eliminate the effect of a competing risk, we treated relapse and death unrelated to aGVHD as competing events. We used Gray's test for comparisons across types of antibiotics administered and applied Fine and Gray's tests for proportional hazards models for the subdistribution of a competing risk to analyze the cumulative incidence of aGVHD [11]. Factors with $P < .10$ in the univariate analysis were included in the multivariate analysis.

Acute myelogenous leukemia and acute lymphoblastic leukemia in the first or second complete remission, chronic myelogenous leukemia in the first or second chronic phase or the accelerated phase, myelodysplastic syndrome with refractory anemia or refractory anemia with ringed sideroblasts, adult T cell leukemia in complete remission, non-Hodgkin lymphoma in complete or partial remission, and aplastic anemia were defined as standard-risk diseases. All other conditions were defined as high risk. The analyzed variables included age, gender, primary malignancy (acute myelogenous leukemia, myelodysplastic syndrome, myeloproliferative disorder, chronic myelogenous leukemia, aplastic anemia, non-Hodgkin

lymphoma, acute lymphoblastic leukemia, adult T cell leukemia, and others), treatment year (≤ 2010 or > 2010), disease risk (standard risk or high risk), source of stem cells (bone marrow, cord blood, or peripheral blood), HLA compatibility (HLA identical or not), conditioning regimen (conventional or reduced-intensity regimen), GVHD prophylaxis (cyclosporine or tacrolimus based), and infection before day 14.

Infection was defined when the following 2 conditions were met. First, the culture of blood obtained during a febrile episode tested positive at least once for bacterial organisms other than common skin contaminants, such as gram-negative bacilli, or certain pathogens such as *Staphylococcus aureus* or fungi. Second, for bacteria that typically colonize the skin, such as coagulase-negative *Staphylococcus*, *Propionibacterium*, viridans group of *Streptococcus*, and non-JK strains of *Corynebacterium*, 2 consecutive positive blood cultures were needed within 72 hours or 1 positive blood culture and 1 positive intravascular catheter tip culture within 72 hours constituted infection.

$P < .05$ were considered statistically significant. All statistical analyses were performed using EZR (Saitama Medical Center, Jichi Medical University, Saitama, Japan), a graphical user interface for R (V.3.3.1; R Foundation for Statistical Computing, Vienna, Austria). More precisely, it is a modified version of R Commander V.2.3-0 that includes statistical functions frequently used in biostatistics [12].

RESULTS

Patient Characteristics

Patient characteristics are displayed in Table 1. The most commonly administered antibiotics in this population were fourth-generation cephalosporins (160 patients, 66%), which are the first-line treatment for neutropenic fever. Piperacillin-tazobactam, an alternative first-line treatment for neutropenic fever, was administered in 115 patients (47%). Quinolone for prophylaxis was prescribed to only 42 patients (17%).

Cumulative Incidence of aGVHD According to the Use of Antibiotics

The cumulative incidences of grades II to IV and III to IV aGVHD were 37.8% (95% confidence interval [CI], 31.6% to 44.2%) and 7.6% (95% CI, 4.6% to 11.5%), respectively. The cumulative incidences of grades II to IV and III to IV aGVHD were 44.3% and 10.5%, respectively, in the population administered fourth-generation cephalosporins, whereas those of grades II to IV and III to IV aGVHD were 26.0% and 1.4%, respectively, in those not administered fourth-generation cephalosporins. In those administered piperacillin-tazobactam, the cumulative incidence of grades II to IV aGVHD was 41.4% (compared with 34.9% in those not administered piperacillin-tazobactam). The results for all 6 evaluated antibiotics are shown in Figure 1.

Univariate Analysis of Factors Associated with Grades II to IV aGVHD

The results of the univariate analysis are displayed in Table 2. The administration of fourth-generation cephalosporins had a significant impact on the occurrence of grades II to IV aGVHD (hazard ratio [HR], 1.98; 95% CI, 1.19 to 3.29; $P = .0087$). In contrast, the administration of piperacillin-tazobactam (HR, .77; 95% CI, .51 to 1.18; $P = .23$) and carbapenems (HR, 1.36; 95% CI, .89 to 2.06; $P = .15$) was not associated with the incidence of grades II to IV aGVHD. There was no significant association between the administration of other antibiotics and

Table 1
Clinical Characteristics of 241 Allogenic HSCT Patients

	Variables	Median or No. of Cases	Range or Percent
Median age, yr (range)		50	(17–66)
Recipient sex			
	Female	108	44.8
	Male	133	55.2
Primary malignancy			
	AML	96	39.8
	MDS/MPD	41	17.0
	CML	7	2.9
	AA	5	2.1
	NHL	37	15.4
	ALL	33	13.7
	ATL	20	8.3
	Others	2	.1
Transplant year			
	2005–2009	119	49.4
	2010–2015	122	50.6
Disease risk			
	Standard risk	140	58.1
	Advanced risk	101	41.9
Graft source			
	Bone marrow	165	68.5
	Cord blood	60	24.9
	Peripheral blood	16	6.6
HLA compatibility			
	HLA identical	128	53.1
	HLA mismatch	113	46.9
Conditioning intensity			
	Myeloablative	121	50.4
	TBI 12 Gy	96	39.8
	TBI 10 Gy	9	3.7
	No TBI	17	7.1
Reduced intensity		120	49.8
	4 Gy	49	20.3
	2 Gy	33	13.7
	No TBI	38	15.8
GVHD prophylaxis			
	TAC-based	199	82.6
	CyA-based	42	17.4
Use of in vivo T cell depletion			
	No	237	98.3
	Yes (antithymocyte globulin)	4	1.7
Antibiotics			
	Fourth-generation cephalosporins	160	66.4
	Glycopeptides	140	58.1
	Piperacillin-tazobactam	115	47.7
	Carbapenems	89	36.9
	Aminoglycosides	74	30.7
	Quinolones as prophylaxis	42	17.4
Media follow-up of survivors, yr (range)		3.74	(.2–10.3)

AML indicates acute myeloid leukemia; MDS, myelodysplastic syndromes; MPD, myeloproliferative disorders; CML, chronic myeloid leukemia; AA, aplastic anemia; NHL, non-Hodgkin lymphoma; ALL, acute lymphoblastic leukemia; ATL, adult T cell leukemia/lymphoma; TBI, total body irradiation; Others, chronic lymphocytic leukemia (n = 1), T cell prolymphocytic leukemia (n = 1); TAC, tacrolimus; CyA, cyclosporine.

aGVHD. Transplant year and donor type were associated with the occurrence of aGVHD, whereas sex, age, conditioning regimen, GVHD prophylaxis, and infection before day 14 were not. Regarding organ-specific aGVHD, the cumulative incidence of gut aGVHD was significantly higher in patients who received fourth-generation cephalosporins than that in those who did not (31% versus 16%, $P = .018$), whereas that of skin or liver aGVHD did not differ significantly between the 2 groups.

We also analyzed the impact of multiple antibiotics. We calculated the HR of grades II to IV aGVHD when 2 types of i.v. antibiotics were administered. In the analysis the combination of fourth-generation cephalosporins with any other antibiotics on aGVHD tended to be associated with the risk of aGVHD. However, no other antibiotic combination was associated with the risk of aGVHD (data not shown). Therefore, we considered

that the multiple antibiotics was not a causative variable for grades II to IV aGVHD.

Multivariate Analysis of Factors Associated with Grades II to IV aGVHD

A multivariable model to determine independent risk factors was constructed using the treatment year, donor type, and the use of fourth-generation cephalosporins. In this model, treatment year (HR, 1.89; 95% CI, 1.20 to 2.98; $P = .006$) and use of fourth-generation cephalosporins (HR, 2.05; 95% CI, 1.24 to 3.39; $P = .0054$) were independently associated with the onset of aGVHD. We additionally performed a separate analysis according to the antibiotic treatment start date and duration (Table 3). The administration of fourth-generation

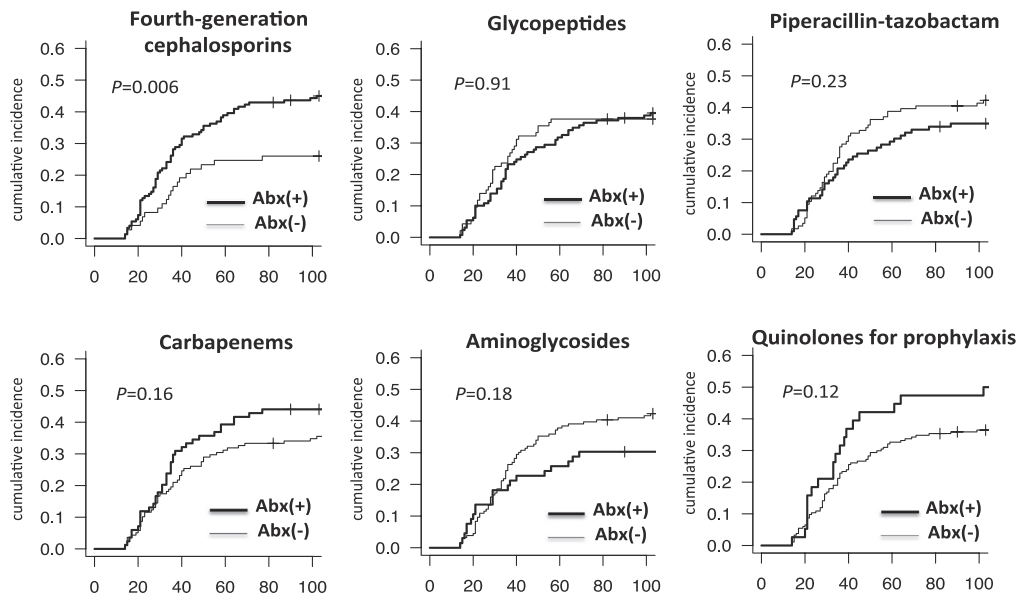


Figure 1. Cumulative incidence of grade II to IV aGVHD compared according to the administration of antibiotics.

Table 2
Univariate and Multivariate Analysis of Factors Associated with Grades II-IV aGVHD

Category	Univariate Analysis		Multivariate Analysis	
	HR (95% CI)	P	HR (95% CI)	P
Age >50 (vs. ≤50), yr	1.30 (.86-1.97)	.22		
Sex, male (vs. female)	.88 (.58-1.33)	.53		
Transplant year, <2010 (vs. ≥2010)	1.92 (1.23-2.99)	.004	1.89 (1.20-2.98)	.047
Disease risk, advanced (vs. standard)	1.15 (.75-1.75)	.52		
Donor type				
HLA-mismatch (vs. HLA match)	1.69 (.99-2.87)	.052	1.53 (.91-2.60)	.089
Cord blood (vs. HLA match)	1.63 (1.01-2.64)	.044	1.28 (.77-2.12)	.290
Conditioning, RIC (vs. MAC)	1.22 (.74-1.70)	.59		
GVHD prophylaxis, CyA-based (vs. TAC-based)	.89 (.50-1.59)	.70		
Infection before day 14	1.11 (.68-1.82)	.67		
Antibiotics				
Fourth-generation cephalosporins	1.98 (1.19-3.29)	.0087	2.05 (1.24-3.39)	.0054
Piperacillin-tazobactam	.77 (.51-1.18)	.23		
Glycopeptides	1.02 (.67-1.57)	.91		
Carbapenems	1.36 (.89-2.06)	.15		
Aminoglycosides	.72 (.44-1.17)	.19		
Quinolones for prophylaxis	1.50 (.91-2.47)	.12		

RIC indicates reduced-intensity conditioning; MAC, myeloablative conditioning.

Table 3
Statistical Analysis of the Start Date and Duration of Fourth-Generation Cephalosporin Treatment

Start Date	HR	The Occurrence of Grades II-IV aGVHD	
		95% CI	P
Control (no use)	1.00		
Days –14 to 0	2.24	1.25-4.01	.007
Days 1-6	1.84	1.05-3.23	.034
Days 7+	2.65	1.27-5.52	.009
Duration			
Control (no use)	1.00		
1-4 days	2.12	1.16-3.85	.014
5-8 days	2.14	1.18-3.88	.012
9+ days	2.01	1.10-3.70	.024

cephalosporins consistently affected the occurrence of aGVHD regardless of the treatment start date and duration.

Impact of Antibiotics on Grades III to IV aGVHD

In univariate analysis, the administration of fourth-generation cephalosporins had a significant impact on the occurrence of grades III to IV aGVHD (HR, 8.03; 95% CI, 1.07 to 60.51; $P = .043$). In contrast, the administration of piperacillin-tazobactam (HR, .060; 95% CI, .22 to 1.63; $P = .32$) and other antibiotics was not associated with the incidence of grades III to IV aGVHD. Disease stage was associated with the onset of aGVHD (HR, 3.11; 95% CI, 1.16 to 8.36; $P = .024$), whereas sex, age, treatment year, conditioning regimen, GVHD prophylaxis, infection before day 14, and donor type were not associated with the onset of grades III to IV aGVHD. A multivariable model was constructed including the use of fourth-generation cephalosporins and disease stage. In this model, the use of fourth-generation cephalosporins (HR, 7.51; 95% CI, 1.00 to 56.21; $P = .06$) tended to increase the risk of grades III to IV aGVHD. Disease stage (HR, 2.91; 95% CI, 1.09 to 7.72; $P = .032$) was independently associated with the onset of grades III to IV aGVHD.

Table 4
Univariate and Multivariate Analysis of Factors Associated with Nonrelapse mortality

Category	Univariate Analysis		Multivariate Analysis	
	HR (95% CI)	P	HR (95% CI)	P
Age >50 (vs. ≤50), yr	1.62 (.88–3.01)	.12		
Sex, male (vs. female)	1.64 (.89–3.00)	.11		
Transplant year, <2010 (vs. ≥2010)	.79 (.43–1.47)	.47		
Disease risk, advanced (vs. standard)	3.09 (1.66–5.76)	.0004	2.89 (1.54–5.45)	.001
Donor type				
HLA mismatch (vs. HLA match)	1.05 (.50–2.18)	.90		
Cord blood (vs. HLA match)	1.02 (.49–2.14)	.95		
Conditioning, RIC (vs. MAC)	1.22 (.67–2.22)	.52		
GVHD prophylaxis, CyA-based (vs. TAC-based)	.60 (.21–1.71)	.34		
Infection before day 14	1.26 (.64–2.48)	.50		
Antibiotics				
Fourth-generation cephalosporins	1.94 (.93–4.03)	.076	1.54 (.73–3.26)	.26
Piperacillin-tazobactam	1.29 (.71–2.35)	.40		
Glycopeptides	1.98 (1.03–3.81)	.042	1.75 (.90–3.39)	.09
Carbapenems	1.31 (.72–2.39)	.38		
Aminoglycosides	.95 (.50–1.83)	.89		
Quinolones as prophylaxis	1.40 (.68–2.90)	.37		

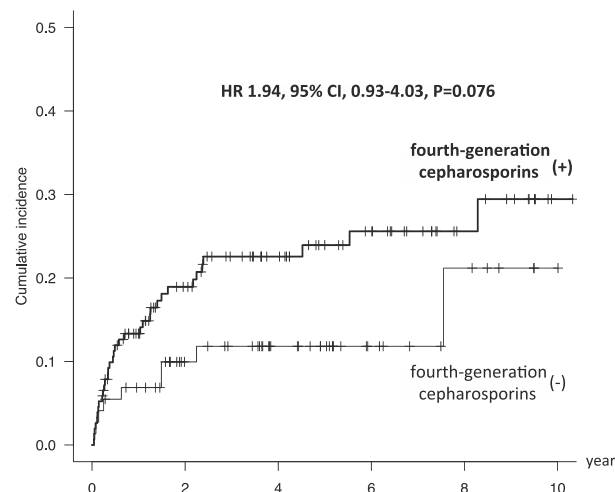


Figure 2. Difference in nonrelapse mortality according to the administration of fourth-generation cephalosporin.

Impact of Antibiotics on Nonrelapse Mortality

In univariate analysis the factors with $P < .10$ included the administration of fourth-generation cephalosporins (HR, 1.94; 95% CI, .93 to 4.03; $P = .076$), glycopeptide (HR, 1.976; 95% CI, 1.03 to

3.81; $P = .042$), and disease stage (HR, 2.681; 95% CI, 1.459 to 4.928; $P = .002$) (Table 4, Figure 2). In multivariable analysis, disease stage was associated with nonrelapse mortality. However, the administration of fourth-generation cephalosporins (HR, 1.54; 95% CI, .73 to 3.31; $P = .26$) and glycopeptides (HR, 1.75; 95% CI, .90 to 3.39, $P = 0.092$) was not associated with nonrelapse mortality.

DISCUSSION

Several reports demonstrated that alterations of intestinal microbiota lead to the onset of various diseases [3,4] because antibiotics change the profiles of intestinal microbiota and consequently may influence the systemic immune status. Thus, we hypothesized that the use and type of antibiotics may affect the occurrence of aGVHD through alterations of the intestinal microbiota.

In this report we elucidated that the administration of fourth-generation cephalosporin consistently influenced the occurrence of aGVHD regardless of the treatment start date. The impact of fourth-generation cephalosporins was not affected by its duration. This result was compatible with the reports that the gut microbiota was rapid with a loss of diversity and a shift in community composition occurring within 3 to 4 days of drug initiation [13,14]. Furthermore, the cumulative incidence of gut aGVHD was significantly higher in patients who received fourth-generation cephalosporins than in those who did not. Previous reports from the United States demonstrated that anaerobic antibiotics such as piperacillin-tazobactam were associated with an increased occurrence of aGVHD, whereas fourth-generation cephalosporins were not [15]. This gap may be related to the following 2 reasons: differences in antibiotic use and racial differences in intestinal microbiota. First, unlike a previous report in which piperacillin-tazobactam was the most commonly administered antibiotic, fourth-generation cephalosporins are the first-line antibiotic for neutropenic fever at our institution and most facilities in Japan [16]. Dethlefsen et al. [14] reported that the repeated disturbance of microbiome by the same antibiotic led to a persistent regime shift. Repetitive exposure to fourth-generation cephalosporins during consolidation therapy may lead to irreversible damage to the intestinal microbiome; as a result, the administration of fourth-generation cephalosporins may have affected the occurrence of aGVHD in our study population. Likewise, the microbiota profile before the transplantation was probably different between our study and the US study's population because the selection of antibiotics was not same during consolidation or earlier therapies. From this viewpoint there was a possibility that the discrepancy of microbiota before the transplantation led to these different results. Second, intestinal microbiota may be the reservoir for antibiotic resistance genes [17]. Hu et al. [18] also reported differences in the numbers and types of antibiotic resistance genes in gut microbiota according to race. Thus, it is probable that the impact of antibiotics on the intestinal microbiota and aGVHD varies among race. Both studies found that the first-line antibiotic used for neutropenic fever was associated with aGVHD. In the present study the use of piperacillin-tazobactam as first-line treatment ($n = 75$) was not associated with the risk of grades II to IV aGVHD (HR, .074; 95% CI, .46 to 1.20; $P = .22$), whereas the use of fourth-generation cephalosporins as first-line treatment ($n = 130$) was significantly associated with the risk of grades II to IV aGVHD (HR, 1.55; 95% CI, 1.00 to 2.41; $P = .049$). This result supports that not the first-line treatment for febrile neutropenia but the fourth-generation cephalosporins itself was a risk factor of aGVHD development.

The study has several limitations. First, it is not certain that the impact of fourth-generation cephalosporins on aGVHD is

mediated through the gut microbiome from these results. To understand this association truly, further studies are required to reveal the impact of antibiotics on the characterization of intestinal microbiota based on 16S metagenomic analyses. Second, at our institution levofloxacin prophylaxis in patients undergoing allogeneic HSCT had not been routinely prescribed. However, oral fluoroquinolones are commonly used as antibacterial prophylaxis after allogeneic HSCT at most institutions. Our results may not be applicable to HSCT using quinolone prophylaxis. Third, the result should be interpreted with caution because of the small sample size and the heterogeneous patient and transplant background. Our findings should be tested in a prospective study in the future.

In conclusion, we demonstrated in this report the following 2 novel insights. First, the administration of fourth-generation cephalosporins consistently affected the occurrence of aGVHD regardless of the treatment start date and duration. Second, the influence of antibiotics on aGVHD may vary within a study population. Fourth-generation cephalosporins are commonly used in cases of febrile neutropenia. The potential impact of fourth-generation cephalosporins on the development of aGVHD may have clinical consequences and requires additional study.

ACKNOWLEDGMENTS

Financial disclosure: The authors have nothing to disclose.

Conflict of interest statement: There are no conflicts of interest to report.

Authorship statement: K.N. performed research, analyzed/interpreted data, and wrote the paper. J.K. designed the study, analyzed/interpreted data, and reviewed the manuscript. M.H., T.K., T.K., K.Y., and A.T.K. reviewed the manuscript. All authors approved the final version submitted.

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